

Engineering Tripos, Part IA, 2023
Paper 3 Electrical and Information Engineering

Solutions

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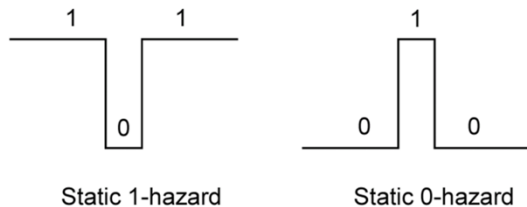
SECTION A

1. (short)

SECTION B

6 (short)

(a) The signal exhibits a momentary state change when it is expected to stay intact primarily due to propagation delays in the circuit. Static 0 and 1 hazard occur when the signal is expected to stay at 0 and 1, respectively.



(b)

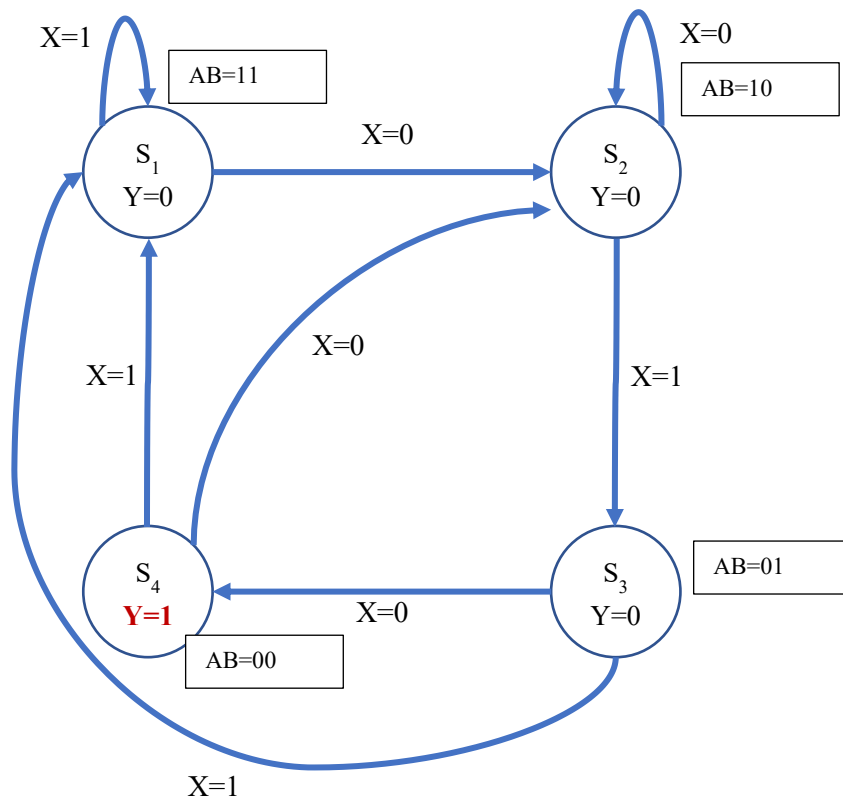
AB \ CD	00	01	11	10
00	1	1	0	0
01	1	1	1	1
11	0	0	1	1
10	0	0	0	0

This is the Karnaugh Map for function F . Clearly, for $D=1$ and $C=0$, $F = \bar{A} + A$ which corresponds to the static 1 hazard situation. To address this hazard, the terms corresponding to the red rectangle in the K-map above, i.e., $\bar{C}D$, could be added to the function F , i.e., $F = \bar{A}\bar{C} + AD + \bar{C}D$

Similarly, we can obtain the complement of the function to see if there is a static 0 hazard, i.e., $\bar{F} = \bar{A}C + A\bar{D} \rightarrow F = \bar{A}\bar{C} + A\bar{D} = \bar{A}\bar{C} \cdot \overline{A\bar{D}} = A \cdot \bar{A}$ when $C=1, D=0$. Hence, there is a static 0 hazard. To mitigate it, the terms corresponding to the blue rectangle in the K-map, i.e., $C\bar{D}$ could be added to the complement of the function F , i.e., $\bar{F} = \bar{A}\bar{C} + AD + C\bar{D}$.

7 (short)

(a) 4 states required as in the state diagram below and therefore 2 bistables will be sufficient to implement the circuit.



(b)

	Current State		Next State		Bistable Inputs			
X	A	B	A	B	J_A	K_A	J_B	K_B
0	1	1	1	0	X	0	X	1
1	1	1	1	1	X	0	X	0
0	1	0	1	0	X	0	0	X
1	1	0	0	1	X	1	1	X
0	0	1	0	0	0	X	X	1
1	0	1	1	1	1	X	X	0
0	0	0	1	0	1	X	0	X
1	0	0	1	1	1	X	1	X

8 (short)

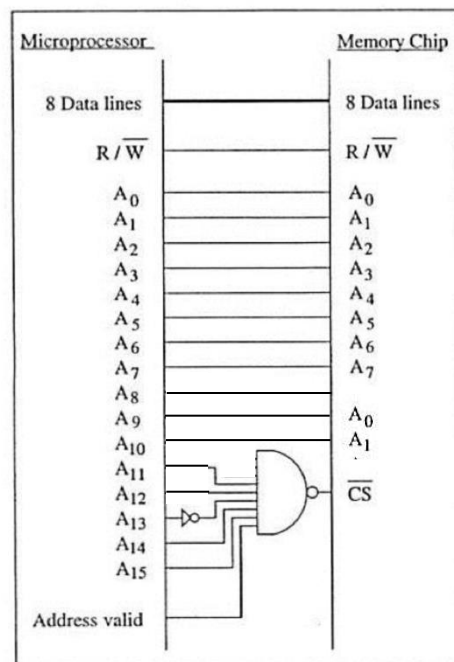
(a) $2^{11} \times 8 \text{ bits} = 2048 \text{ bits} = 2 \text{ KBytes}$ is the capacity of the memory chip.

(b)

$D800_H \rightarrow 1101 \mid 1000 \mid 0000 \mid 0000$

$DFFF_H \rightarrow 1101 \mid 1111 \mid 1111 \mid 1111$
 5-bits CS 11-bits the address space

$$\rightarrow \overline{CS} = \overline{A_{15}A_{14}A_{13}A_{12}A_{11}}$$



9 (long)

(a)

Current State				Next State				J ₃	K ₃	J ₂	K ₂	J ₁	K ₁	J ₀	K ₀
Q ₃	Q ₂	Q ₁	Q ₀	Q ₃ ⁺	Q ₂ ⁺	Q ₁ ⁺	Q ₀ ⁺								
1	1	1	1	1	1	1	0	X	0	X	0	X	0	X	1
1	1	1	0	1	1	0	1	X	0	X	0	X	1	1	X
1	1	0	1	1	1	0	0	X	0	X	0	0	X	X	1
1	1	0	0	1	0	1	1	X	0	X	1	1	X	1	X
1	0	1	1	1	0	1	0	X	0	0	X	X	0	X	1
1	0	1	0	1	0	0	1	X	0	0	X	X	1	1	X
1	0	0	1	1	0	0	0	X	0	0	X	0	X	X	1
1	0	0	0	0	1	1	1	X	1	1	X	1	X	1	X
0	1	1	1	0	1	1	0	0	X	X	0	X	0	X	1
0	1	1	0	0	1	0	1	0	X	X	0	X	1	1	X
0	1	0	1	0	1	0	0	0	X	X	0	0	X	X	1
0	1	0	0	0	0	1	1	0	X	X	1	1	X	1	X
0	0	1	1	0	0	1	0	0	X	0	X	X	0	X	1
0	0	1	0	0	0	0	1	0	X	0	X	X	1	1	X
0	0	0	1	0	0	0	0	0	X	0	X	0	X	X	1
0	0	0	0	1	1	1	1	1	X	1	X	1	X	1	X

(b)

J₀

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	1	X	X	1
	01	1	X	X	1
	11	1	X	X	1
	10	1	X	X	1

$$J_0 = 1$$

K₀

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	X	1	1	X
	01	X	X	1	X
	11	X	1	1	X
	10	X	1	1	X

$$K_0 = 1$$

J₁

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	1	0	X	X
	01	1	0	X	X
	11	1	0	X	X
	10	1	0	X	X

K₁

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	X	X	0	1
	01	X	X	0	1
	11	X	X	0	1
	10	X	X	0	1

$$J_1 = \overline{Q_0}$$

$$K_1 = \overline{Q_0}$$

J₂

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	1	0	0	0
	01	X	X	X	X
	11	X	X	X	X
	10	1	0	0	0

$$J_2 = \overline{Q_1} \cdot \overline{Q_0}$$

K₂

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	X	X	X	X
	01	1	0	0	0
	11	1	0	0	0
	10	X	X	X	X

$$K_2 = \overline{Q_1} \cdot \overline{Q_0}$$

J₃

		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	1	0	0	0
	01	0	0	0	0
	11	X	X	X	X
	10	X	X	X	X

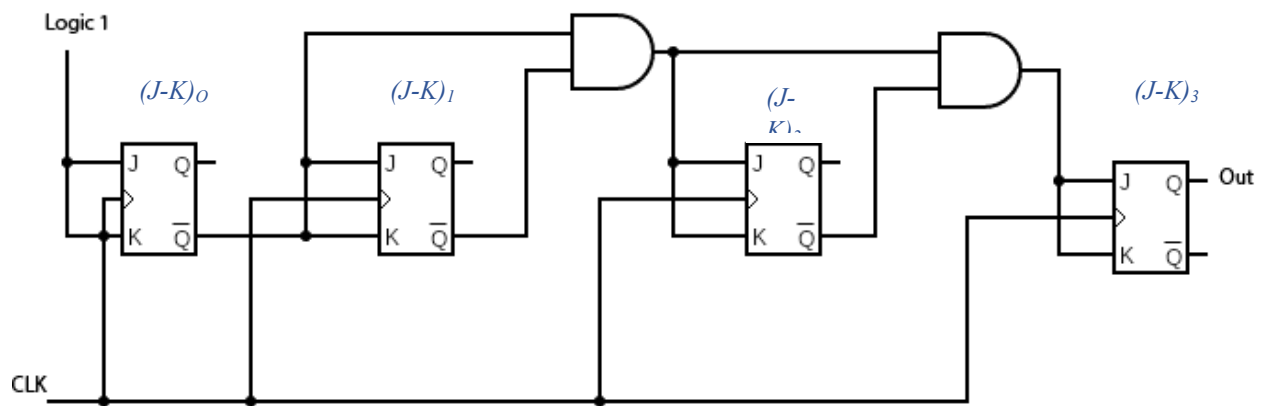
$$J_3 = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}$$

K₃

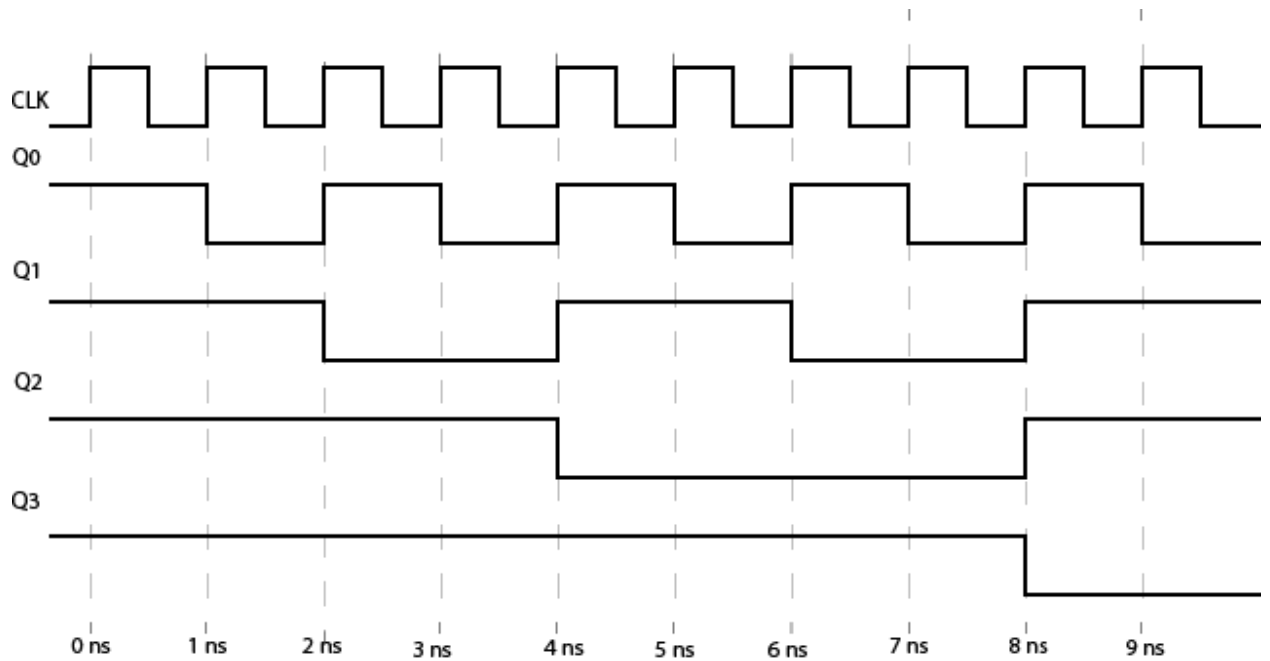
		Q ₁ Q ₀			
		00	01	11	10
Q ₃ Q ₂	00	X	X	X	X
	01	X	X	X	X
	11	0	0	0	0
	10	1	0	0	0

$$K_3 = \overline{Q_2} \cdot \overline{Q_1} \cdot \overline{Q_0}$$

(c)

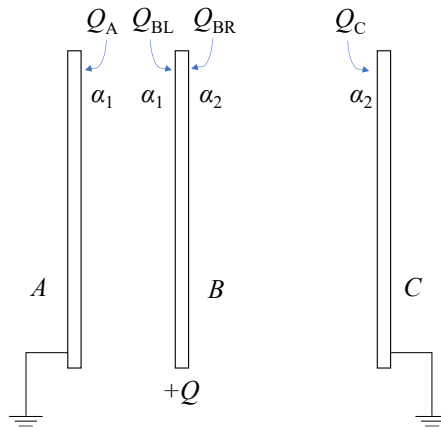


(d)



SECTION C

10 (short)



$$Q_{BL} + Q_{BR} = Q$$

$$Q_A = -Q_{BL}; Q_C = -Q_{BR}$$

$$(a) D_A * A = Q_A = \alpha_1 * A$$

$$\epsilon_r * E_{AB} = \alpha_1; \epsilon_r * E_{BC} = \alpha_2$$

$$V_{AB} = E_{AB} * d_1; V_{BC} = E_{BC} * d_2;$$

$$E_{AB} * d_1 = E_{BC} * d_2;$$

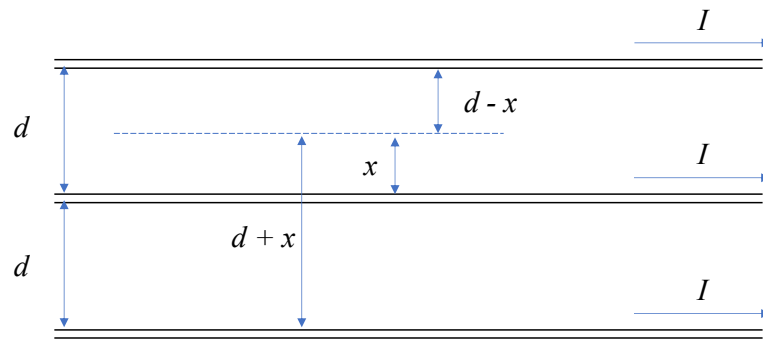
$$\frac{\alpha_1}{\alpha_2} = \frac{d_2}{d_1} = 2$$

$$Q_A = Q_{BL} = -2.0 \times 10^{-7} \text{ C}$$

$$Q_C = Q_{BR} = -1.0 \times 10^{-7} \text{ C}$$

$$(b) V_{AB} = V_{BC} = E_{AB} * d_1 = (\alpha_1 * d_1) / \epsilon_r = 2.3 \times 10^3 \text{ V}$$

11 (short)



The location that $B=0$ must be within the same plane with the three wires. Assume the location that $B=0$ is x away from the central wire, and so the total B can be given by:

$$B_{\text{total}} = B_1 + B_2 + B_3$$

$$B_{\text{total}} = \frac{\mu_0 I}{2\pi x} + \frac{\mu_0 I}{2\pi(d+x)} - \frac{\mu_0 I}{2\pi(d-x)}$$

$$\text{Let } B_{\text{total}} = 0, x = \pm \frac{1}{\sqrt{3}}d = \pm 0.5773 d;$$

$x = 0$ would also be a solution.

12 (long)

(a) Electric field:

At $R=1.0$ cm, inside the conducting sphere, so $E=0$;

At $R=3.0$ cm,

$$E = \frac{Q}{4\pi\epsilon_0 R^2} = 3 \times 10^5 \text{ V/m}$$

At $R=6.0$ cm, outside the outer surface that got connected to earth,

So $E=0$.

(b) Capacitor of the concentric spheres:

$$Q = CV;$$

$$C = \frac{Q}{V}$$

$$V = \int \frac{Q}{4\pi\epsilon_0 R^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$C = \frac{4\pi\epsilon_0}{\frac{1}{R_1} - \frac{1}{R_2}} = 4.49 \times 10^{-12} \text{ F}$$

(c) Total electrostatic energy:

$$W = \frac{1}{2} CV^2 = \frac{1}{2} QV = 1.01 \times 10^{-4} \text{ J}$$